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Vision Transformers (*ViT*) and Their Modern Variants in Image Classification

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Introduction / Motivation to the Topic

Vision Transformers (ViT) have recently emerged as one of the most popular and widely cited approaches in computer vision. Inspired by Transformer architectures from natural language processing, ViTs have demonstrated highly competitive performance in image classification tasks, sometimes surpassing traditional convolutional neural networks (CNNs).

This research project aims to investigate ViTs and their modern variants, analyzing their architectures, comparing their performance, and running practical implementations on small datasets in Google Colab. The motivation for selecting this topic is the combination of high academic relevance, strong citation counts, and practical feasibility for a student project.

Topic Description

Vision Transformers (ViT) apply the self-attention mechanism of Transformers to image data. Images are divided into patches, each patch is flattened and embedded, and the sequence of patch embeddings is processed by a standard Transformer encoder. The output class token is then used for classification.

Modern variants of ViT include:

- **DeiT (Data-efficient Image Transformers)** – a lighter, more efficient version of ViT
- **Swin Transformer** – hierarchical structure for better local representation
- **MobileViT** – optimized for mobile and edge devices

These variants allow flexibility in implementation and experimentation, making the topic highly suitable for research and code execution in Google Colab.

Research Goal

Vision Transformers (ViT) apply the self-attention mechanism of Transformers to image data. Images are divided into patches, each patch is flattened and embedded, and the sequence of patch embeddings is processed by a standard Transformer encoder. The output class token is then used for classification.

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Main Survey Paper

Paper: *Transformers in Vision: A Survey*
Authors: Khan, Naseer, Hayat, Zamir, Shahbaz Khan & Shah
Year: 2021
Citations: ~2,200
Link: https://arxiv.org/abs/2101.01169

Description / Justification

This paper is selected as the main survey because it provides a comprehensive overview of Vision Transformers (ViTs) and their applications in computer vision, including:

- Classification
- Detection
- Segmentation
- Video and multimodal tasks

It also discusses architecture, variants, advantages, limitations, and future directions. The paper is highly cited (~2,200 citations), making it academically credible, and is well-structured and readable, which simplifies understanding and summarization.

Supporting Papers

1. *A Survey on Efficient Vision Transformers: Algorithms, Techniques, and Performance Benchmarking (2023)* https://arxiv.org/abs/2309.02031
2. *Vision Transformers for Image Classification: A Comparative Survey (2025)* https://doi.org/10.3390/technologies13010032
3. *Exploring the Synergies of Hybrid CNNs and ViTs Architectures for Computer Vision: A survey (2024)* https://arxiv.org/abs/2402.02941

Introduction

Vision Transformers (ViTs) have emerged as a groundbreaking approach in computer vision by adapting the transformer architecture, originally designed for natural language processing, to image data. Unlike convolutional neural networks (CNNs), which rely on local receptive fields and convolutional kernels, ViTs process images by dividing them into fixed-size patches and using self-attention to model global relationships among patches.

This project aims to explore the structure, variants, and applications of ViTs, and to perform practical experiments using pre-trained models on small datasets in Google Colab. The goal is to provide a comprehensive analysis of ViTs and compare their performance, efficiency, and usability for real-world tasks.

Why ViTs are important:

- *They achieve competitive or superior performance on image classification benchmarks.*
- *They provide a **unified architecture** for multiple vision tasks (classification, detection, segmentation).*
- *They allow **flexible scaling** (tiny models to huge models like ViT-Large).*

Motivation

The topic was chosen due to a combination of **academic relevance, practicality, and research potential**:

1. **High impact**: ViT is widely cited (~2,200 citations), indicating strong influence in the research community.
2. **Colab-friendly experimentation**: Pre-trained models allow hands-on demos without heavy computation.
3. **Comparison opportunity**: Multiple variants exist with different architectures, making it easy to analyze pros, cons, and efficiency.

4. **Pedagogical value**: Easy to explain visually (patch embedding, attention maps) and conceptually simple compared to more complex AI models (GANs, RL).

Background

Transformer Basics

- **Self-Attention:** Calculates a weighted sum of all input tokens, where weights depend on similarity between tokens.
- **Multi-Head Attention:** Captures multiple types of relationships in parallel.
- **Feed-Forward Networks (MLPs):** Process the attention output for each token individually.
- **Positional Encoding:** Since transformers are permutation-invariant, position information must be added to tokens.

Vision Transformers (ViT)

Steps in a ViT model:

1. **Image Patchification:** Split image into fixed-size patches (e.g., 16×16 pixels).
2. **Flattening & Linear Projection:** Each patch is flattened into a vector and projected to a high-dimensional embedding.
3. **Add Positional Encoding:** Ensures the model understands spatial arrangement.
4. **Transformer Encoder:** Several layers of self-attention + MLPs process the patch embeddings.
5. **Classification Head:** Use the special class token for final prediction.

Vision Transformer Variants

Variant	*Key Idea*	*Strengths*	*Practical Notes*
ViT (original)	*Standard transformer on patches*	*Strong on large datasets*	*Needs pre-trained weights*
DeiT	*Data-efficient, uses distillation*	*Can train on smaller datasets*	*Ideal for Colab*
Swin Transformer	*Hierarchical, multi-scale*	*Good for detection/segmentation*	*Slightly more complex*
MobileViT	*Lightweight & mobile-friendly*	*Efficient, small memory*	*Runs on CPU/low-GPU easily*

Practical Implementation

Goal: Show a working example without training from scratch.

Steps:

1. Load pre-trained model using timm.
2. Load small dataset (CIFAR-10 or MNIST).
3. Preprocess images to 224×224 and normalize.
4. Pass through the model and observe outputs.

Example Python snippet:

```
!pip install timm
import timm, torch
from torchvision import datasets, transforms

model = timm.create_model('vit_tiny_patch16_224', pretrained=True)
model.eval()

transform = transforms.Compose([transforms.Resize((224,224)),
transforms.ToTensor()])
```

```
dataset = datasets.CIFAR10(root='./data', train=False, download=True,
transform=transform)
image, label = dataset[0]
image = image.unsqueeze(0)
output = model(image)
print("Output logits:", output)
```

Comparison and Analysis

Metrics to compare ViT variants:

- Accuracy on benchmark datasets (ImageNet, CIFAR-10)
- Number of parameters
- Inference speed / latency
- Resource requirement (CPU/GPU)
- Suitability for deployment (mobile, edge, desktop)

Initial Observations (from surveys):

- DeiT is ideal for small-scale experiments (low compute, high accuracy).
- Swin Transformer improves performance for detection/segmentation.
- MobileViT balances accuracy with efficiency.
- Original ViT requires large datasets and is heavy for small-scale demos.

Comparison and Evaluation of Vision Transformer Methods

Evaluation Criteria

To compare Vision Transformer (ViT) models and their variants in a structured and fair way, the following evaluation criteria are considered:

1. **Classification Accuracy** – Performance on standard image classification benchmarks such as ImageNet and CIFAR-10.
2. **Model Size (Parameters)** – Number of parameters, which affects memory usage and deployment feasibility.
3. **Computational Cost** – Training and inference complexity, including FLOPs and GPU/CPU requirements.
4. **Data Efficiency** – Ability to perform well with limited training data.
5. **Practical Usability** – Ease of implementation, availability of pre-trained models, and compatibility with Google Colab.

These criteria are commonly used in the surveyed literature and provide a balanced view of both theoretical performance and practical applicability.

Compared Methods

Based on the selected survey paper and supporting references, the following methods are compared:

- **Vision Transformer (ViT)**
- **Data-efficient Image Transformer (DeiT)**
- **Swin Transformer**
- **MobileViT**

These models represent different design philosophies within the Vision Transformer family.

Model-wise Analysis

1. Vision Transformer (ViT)

The original ViT model applies a standard Transformer encoder directly to image patches.

Strengths:

- Simple and elegant architecture
- Strong performance when trained on large-scale datasets
- Highly scalable

Weaknesses:

- Requires very large datasets for effective training
- Computationally expensive
- Less suitable for small-scale or resource-limited environments

Evaluation Summary:
ViT serves as a strong baseline but is not ideal for small datasets or limited computational resources without pre-training.

2. DeiT (Data-efficient Image Transformer)

DeiT introduces **knowledge distillation** to make ViT training more data-efficient.

Strengths:

- Performs well with smaller datasets
- Lower computational cost than original ViT
- Designed explicitly for practical training scenarios

Weaknesses:

- Slightly lower peak accuracy than large ViT models

- Still heavier than CNN-based lightweight models

Evaluation Summary:
DeiT is one of the most practical Vision Transformer variants and is well-suited for student-level experimentation and Google Colab execution.

3. Swin Transformer

Swin Transformer introduces a **hierarchical architecture** and **shifted windows** to capture local and global information efficiently.

Strengths:

- Excellent performance on both classification and downstream tasks
- Hierarchical feature representation similar to CNNs
- Strong scalability

Weaknesses:

- More complex architecture
- Harder to explain and implement compared to ViT and DeiT
- Higher implementation overhead

Evaluation Summary:
Swin Transformer achieves strong performance but sacrifices architectural simplicity, making it less ideal for introductory research demonstrations.

4. MobileViT

MobileViT is designed for **mobile and edge devices**, combining CNN layers with Transformer blocks.

Strengths:

- Lightweight and efficient
- Low memory usage

- Fast inference on CPU

Weaknesses:

- Lower accuracy compared to larger ViT variants
- Less flexible for high-performance tasks

Evaluation Summary:
MobileViT is highly suitable for deployment-focused scenarios and constrained environments, though with reduced accuracy.

Comparative Table

Model	Accuracy	Parameters	Compute Cost	Data Efficiency	Colab Suitability
ViT	High (large data)	Very High	High	Low	Medium
DeiT	High	Medium	Medium	High	High
Swin Transformer	Very High	High	High	Medium	Medium
MobileViT	Medium	Low	Low	High	Very High

Comparative Table of Models

Values are comparative and based on survey analysis rather than final experimental results.

LLM-Assisted Insights

Using LLM-based analysis (e.g., summarization and comparison of survey papers), several patterns emerge:

- **Architectural simplicity** strongly correlates with ease of experimentation.

- **Hybrid designs** (CNN + Transformer) offer a good trade-off between accuracy and efficiency.
- **Data efficiency techniques** (e.g., distillation in DeiT) are crucial for real-world usability.

LLMs help accelerate literature analysis and highlight common evaluation metrics, but **final judgments require human interpretation**, especially regarding practical constraints.

Personal Analysis and Interpretation

- **DeiT** offers the best balance between performance and feasibility.
- **ViT** is conceptually important but less practical without large datasets.
- **Swin Transformer** is powerful but complex, making it better suited for advanced research.
- **MobileViT** is ideal for lightweight deployment scenarios.

For a university research project with limited resources, **data-efficient and lightweight Vision Transformers provide the most value**.

Abstract

Vision Transformers (ViTs) have recently emerged as a powerful alternative to convolutional neural networks for image classification tasks by leveraging the self-attention mechanism of Transformer architectures. This research investigates the fundamental structure of Vision Transformers and analyzes several prominent variants, including ViT, DeiT, Swin Transformer, and MobileViT.

Using survey-based analysis supported by LLM-assisted literature review and practical experimentation in Google Colab, the models are evaluated based on accuracy, computational cost, data efficiency, and practical usability. The results indicate that while original ViT models achieve strong performance on large-scale datasets, data-efficient and lightweight variants such as DeiT and MobileViT provide a better balance between performance and feasibility in resource-constrained environments. This study highlights the importance of architectural design choices in determining the suitability of Vision Transformer models for real-world and educational applications.

Conclusion

This research explored Vision Transformers and their modern variants through a combination of literature review, LLM-assisted analysis, and practical evaluation. The comparison of different models demonstrates that Vision Transformers are not a single unified solution but rather a diverse family of architectures designed to address various computational and data constraints.

The analysis shows that original ViT models, while conceptually elegant and highly scalable, are less suitable for small-scale experiments without extensive pre-training. In contrast, data-efficient approaches such as DeiT significantly improve usability by reducing training requirements, while lightweight hybrid models like MobileViT enable deployment in constrained environments. Swin Transformer achieves strong performance through hierarchical representation but introduces additional architectural complexity.

Overall, this study concludes that **model efficiency and practicality are critical factors** when selecting Vision Transformer architectures for real-world use and academic experimentation. Future work will focus on deeper experimental evaluation, fine-tuning on custom datasets, and exploring optimization techniques to further improve performance and efficiency.